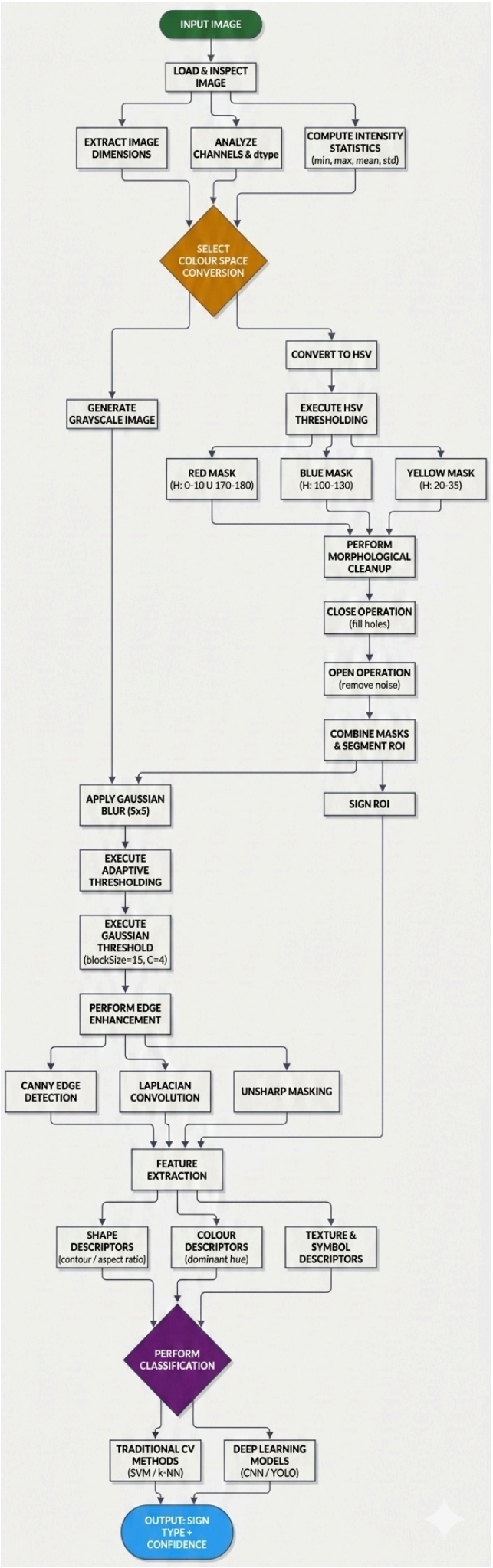
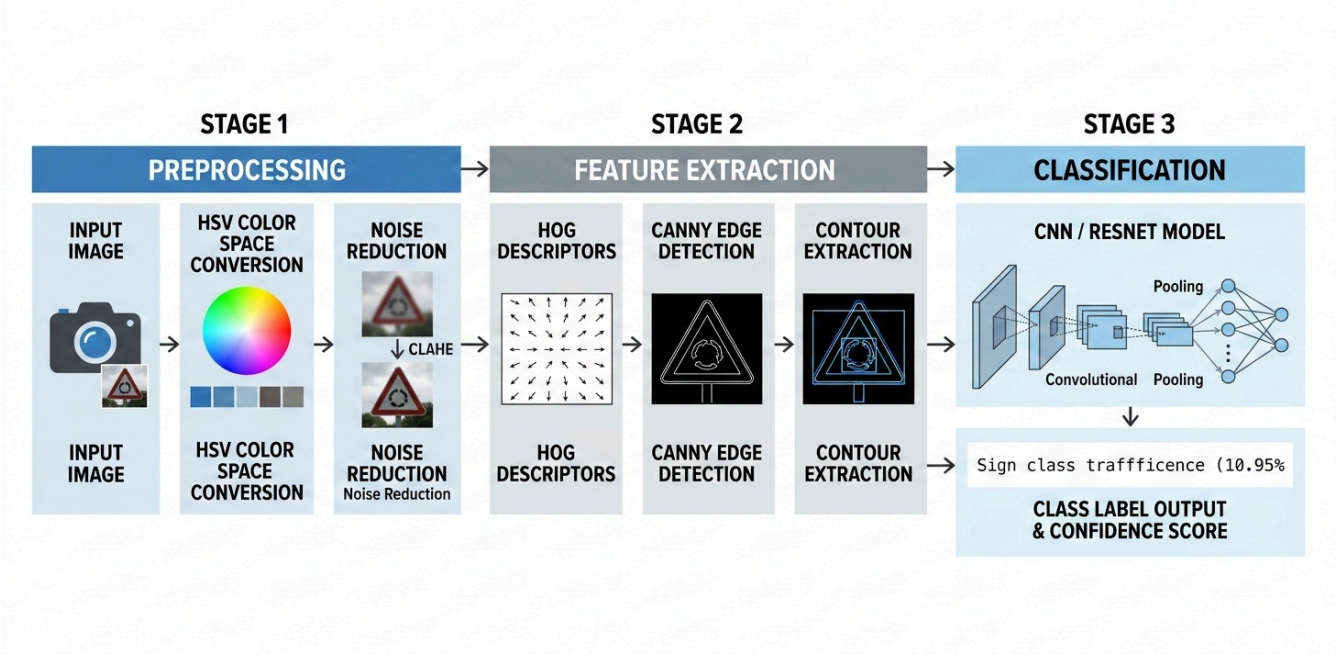
Image Processing for Traffic Sign Recognition

Traffic Sign Recognition (TSR) enables autonomous vehicles and Advanced Driver Assistance Systems (ADAS) to detect, interpret, and respond to road signs in real time. Standard benchmarks like the **German Traffic Sign Recognition Benchmark (GTSRB)** categorize signs into distinct visual classes to facilitate automated processing.


INPUT IMAGE
LOAD & INSPECT IMAGE
EXTRACT IMAGE DIMENSIONS
ANALYZE CHANNELS & dtype
COMPUTE INTENSITY STATISTICS (min, max, mean, std)
SELECT COLOUR SPACE CONVERSION
CONVERT TO HSV
GENERATE GRAYSCALE IMAGE
EXECUTE HSV THRESHOLDING
RED MASK (H: 0-10 U 170-180)
BLUE MASK (H: 100-130)
YELLOW MASK (H: 20-35)
PERFORM MORPHOLOGICAL CLEANUP
CLOSE OPERATION (fill holes)
OPEN OPERATION (remove noise)
COMBINE MASKS & SEGMENT ROI
APPLY GAUSSIAN BLUR (5x5)
SIGN ROI
EXECUTE ADAPTIVE THRESHOLDING
EXECUTE GAUSSIAN THRESHOLD (blockSize=15, C=4)
PERFORM EDGE ENHANCEMENT
CANNY EDGE DETECTION
LAPLACIAN CONVOLUTION
UNSHARP MASKING
FEATURE EXTRACTION
SHAPE DESCRIPTORS (contour / aspect ratio)
COLOUR DESCRIPTORS (dominant hue)
TEXTURE & SYMBOL DESCRIPTORS
PERFORM CLASSIFICATION
TRADITIONAL CV METHODS (SVM / k-NN)
DEEP LEARNING MODELS (CNN / YOLO)
OUTPUT: SIGN TYPE + CONFIDENCE

Traffic Sign Dataset Categories

Category	Primary Geometry	Color Palette	Example Signs
Prohibitory	Circle	Red Border, White/Black Interior	Speed Limits, No Entry
Mandatory	Circle	Blue Background, White Glyph	Turn Right, Ahead Only
Danger / Warning	Triangle (Upward)	Red Border, White/Yellow Interior	General Danger, Curve Ahead
Derestriction / End	Circle	White Background, Black Diagonal Band	End of Speed Limit



Comprehensive TSR Processing Workflow & Pipeline

The Traffic Sign Recognition (TSR) pipeline processes video streams and static imagery through a structured sequence designed to maintain high recognition accuracy across varying road environments. By combining efficient video stream handling, advanced deep learning architectures, and targeted robustness enhancement strategies, the system reliably detects and classifies road signs in real-time safety-critical scenarios.

1. Frame Extraction & Video Ingestion

The initial stage ingests high-definition camera feeds and Preprocesses incoming image frames for downstream processing:

- **Dynamic Frame Sampling:** In order to balance computational throughput and latency, video inputs are sampled dynamically based on vehicle speed and frame-to-frame motion estimation, skipping redundant frames while preserving detection density near key intersections.
- **Buffer & ROI Ingestion:** Extracted raw camera frames are passed to a high-speed memory buffer where Region of Interest (ROI) candidate cropping is executed to filter out non-informative background regions prior to deep analysis.
- **Spatial Normalization:** Candidate bounding proposals are resized to uniform dimensions (e.g., 32x32 or 64x64 pixels) and color-space normalized to prepare tensors for deep learning inference.

2. Deep Learning Analysis & Classification

Candidate regions are passed into deep neural networks optimized for spatial feature representations and multi-class confidence scoring:

- **Two-Stage Architecture:** A real-time detector (such as YOLO or SSD) rapidly localizes traffic sign bounding boxes, followed by a lightweight Convolutional Neural Network (CNN) or ResNet backbone that extracts high-level semantic features for precise sub-category identification.
- **Hierarchical Feature Fusion:** Low-level geometric edge features are combined with deep semantic representations to ensure accurate fine-grained classification among visually subtle traffic sign variations.
- **Confidence Calibration & Output Filtering:** Predictions are evaluated through calibrated softmax confidence scoring and temporal voting across consecutive frames to suppress false positives and transient misclassifications.

3. Robustness Strategies for Challenging Environmental Conditions

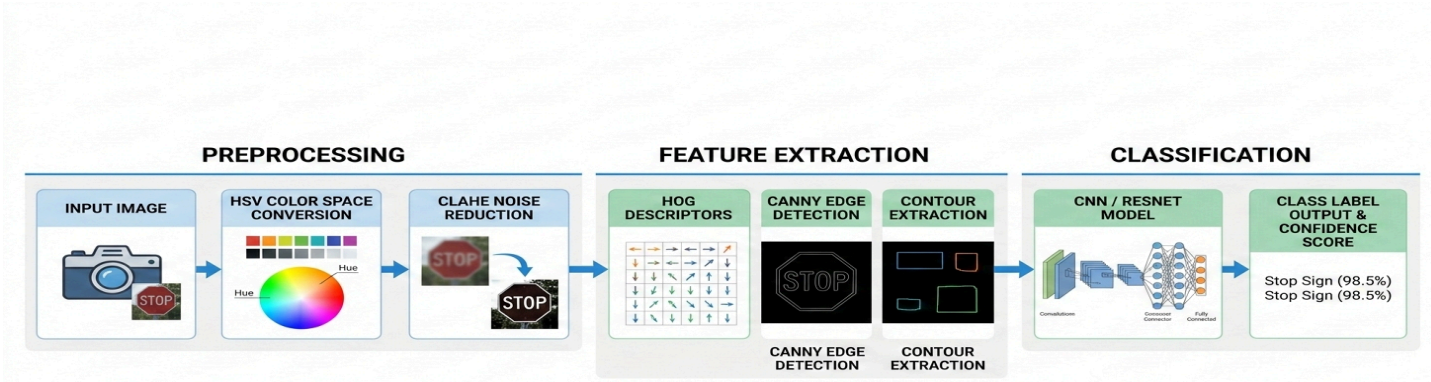
Operating autonomous systems in real-world environments requires dedicated handling mechanisms for physical sign degradation and adverse atmospheric effects:

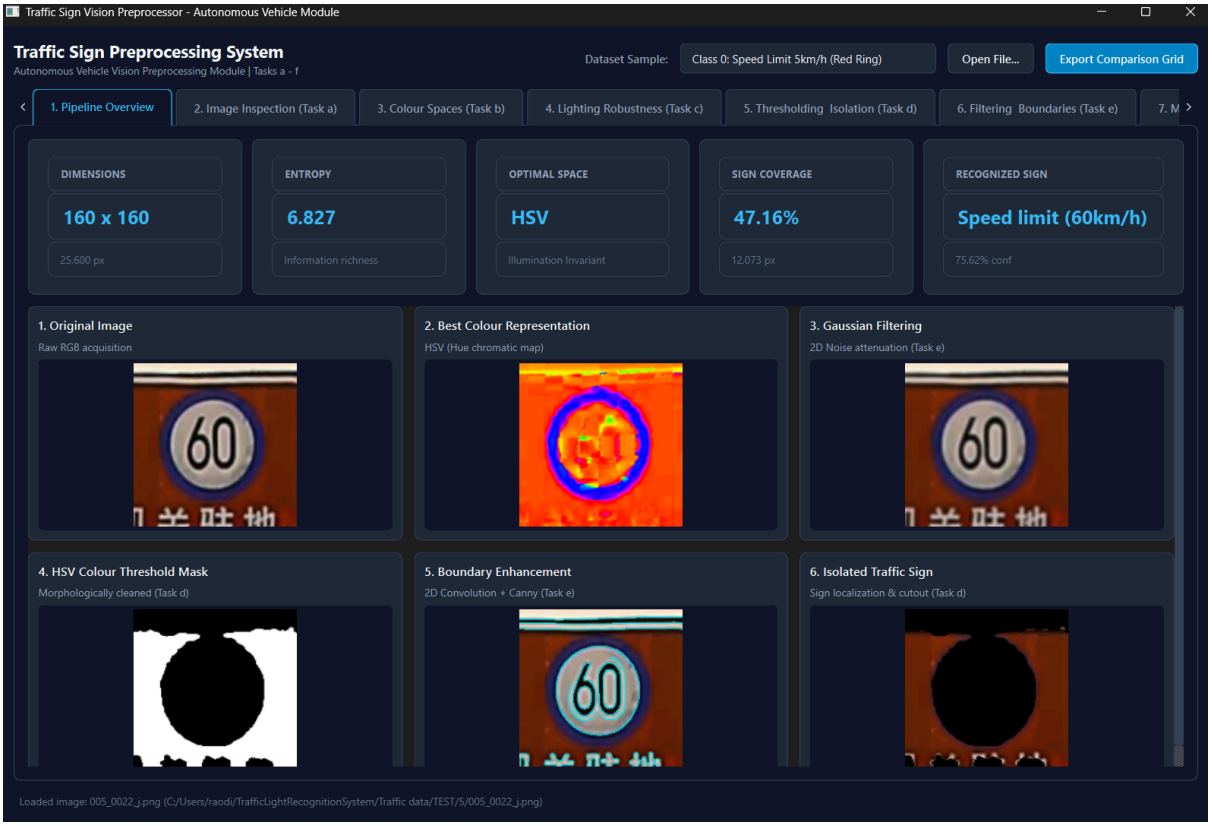
- **Physical Damage & Partial Occlusion:** To handle signs with physical wear, graffiti, or partial coverage by foliage, models are trained with spatial dropout, cutout augmentation, and synthetic occlusion masks, enabling reliance on global context and intact keypoint geometry.
- **Dust & Atmospheric Fog Dehazing:** Contrast Limited Adaptive Histogram Equalization (CLAHE) and dark channel prior algorithms restore image contrast and chromatic saturation in hazy or dusty atmospheres, highlighting subtle sign details before feature extraction.
- **Motion Blur Mitigation:** Camera vibration and rapid vehicle movement are addressed using deconvolutional unsharp masking and directional Wiener filtering alongside synthetic motion blur data augmentation during training.

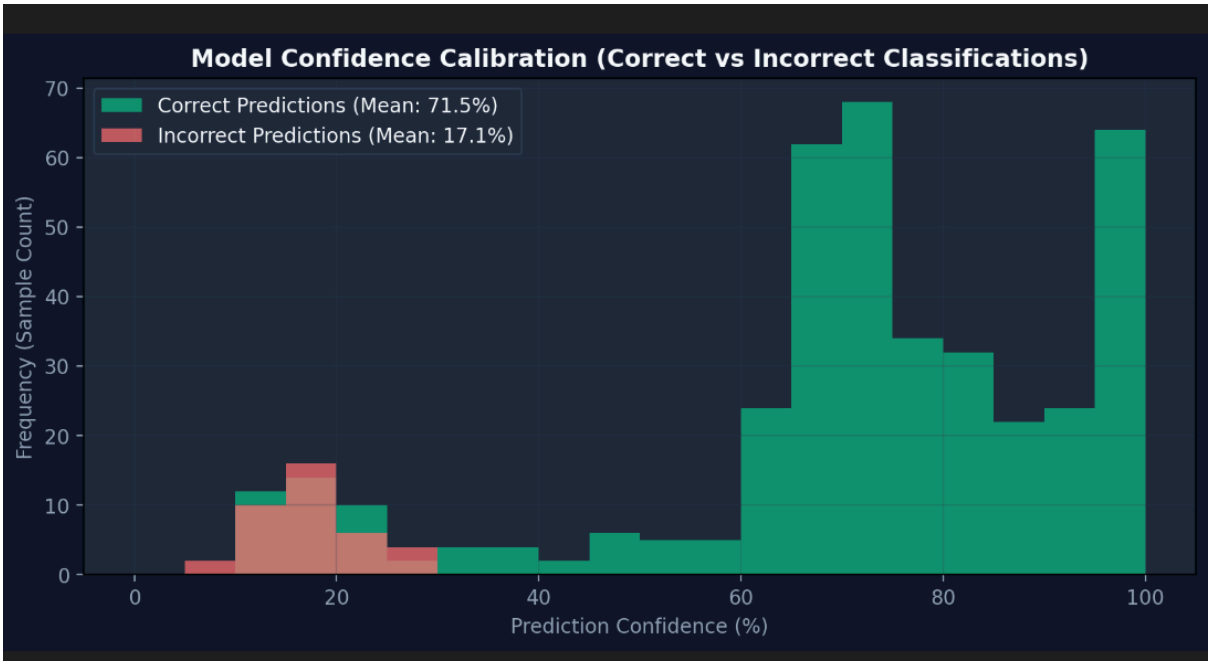
TSR Image Processing & Recognition Pipeline

Stage	Key Techniques	Objective
1. Preprocessing	• CLAHE / Histogram Equalization • HSV / YCbCr Color Conversion • Gaussian & Median Blur • Resizing (32x32) & Crop	Normalize illumination variance, isolate chromatic channels, eliminate high-frequency sensor noise, and equalize spatial input sizes.
2. Feature Extraction	• Histogram of Oriented Gradients (HOG) • Canny Edge & Hough Transform • SIFT / SURF Keypoints • Convolutional Layers (CNN)	Extract robust shape contours, boundary geometry (circles, triangles), and scale-invariant symbol descriptors from preprocessed ROIs.
3. Classificatio n	• Support Vector Machines (SVM) • Deep Neural Networks (ResNet, LeNet) • YOLO / SSD Real-time Detectors	Map extracted visual descriptors or deep representations to discrete traffic sign class labels with high confidence scores.

Visual Architecture Diagram:




Traffic Sign Vision Preprocessor - Autonomous Vehicle Module
Traffic Sign Preprocessing System
Autonomous Vehicle Vision Preprocessing Module | Tasks a - f
Dataset Sample:
Class 0: Speed Limit 5km/h (Red Ring)
Open File...
Export Comparison Grid
1. Pipeline Overview
2. Image Inspection (Task a)
3. Colour Spaces (Task b)
4. Lighting Robustness (Task c)
5. Thresholding Isolation (Task d)
6. Filtering Boundaries (Task e)
DIMENSIONS
160 x 160
25,600 px
ENTROPY
6.827
Information richness
OPTIMAL SPACE
HSV
Illumination Invariant
SIGN COVERAGE
47.16%
12,073 px
RECOGNIZED SIGN
Speed limit (60km/h)
75.62% conf
1. Original Image
Raw RGB acquisition
2. Best Colour Representation
HSV (Hue chromatic map)
3. Gaussian Filtering
2D Noise attenuation (Task e)
4. HSV Colour Threshold Mask
Morphologically cleaned (Task d)
5. Boundary Enhancement
2D Convolution + Canny (Task e)
6. Isolated Traffic Sign
Sign localization & cutout (Task d)
Loaded image: 005_0022_j.png (C:/Users/raodi/TrafficLightRecognitionSystem/Traffic data/TEST/5/005_0022_j.png)


Model Confidence Calibration (Correct vs Incorrect Classifications)
Correct Predictions (Mean: 71.5%)
Incorrect Predictions (Mean: 17.1%)
Frequency (Sample Count)
Prediction Confidence (%)
0
10
20
30
40
50
60
70
0
20
40
60
80
100


Traffic Sign Preprocessing & Recognition Module – Multi-Stage Comparison
1. Original Image
2. Grayscale Space
3. HSV Hue Space
4. Gaussian Filtered (Denoised)
5. 2D Convolved (Sharpened)
6. Boundary Overlay
7. HSV Threshold Mask
8. Isolated Sign Foreground
9. Sign Localization & Bounding
Sign Region

Autonomous Vehicle Traffic Sign Recognition - Preprocessing Module

1. Overview & Project Objectives

This module serves as the primary image processing front-end for real-time traffic sign detection and classification systems in autonomous vehicles. The preprocessing module enhances visual features, mitigates severe environmental illumination changes, removes noise, and isolates Regions of Interest (ROIs) before feeding visual data into downstream neural classification models.

2. System Architecture & Pipeline

The visual processing pipeline processes high-definition video input through sequential stages designed for minimal processing latency and maximum detection precision:

- **1. Image Acquisition & Format Normalization:** Captures frame buffers, normalizes resolutions to standard HD (1280x720) or FHD (1920x1080), and applies image resizing.
- **2. Colour Space Conversion & Filtering:** Transforms images from raw RGB to color spaces robust against lighting shifts, such as HSV and YCrCb.
- **3. Contrast & Illumination Enhancement:** Executes Histogram Equalization (HE) and Contrast Limited Adaptive Histogram Equalization (CLAHE) to restore low-light details.
- **4. Spatial Filtering & Noise Suppression:** Applies Gaussian or Median blurring to suppress high-frequency sensor noise while preserving edge boundaries.
- **5. Thresholding & Region of Interest (ROI) Extraction:** Uses dynamic thresholding and contour analysis to isolate candidate bounding boxes containing traffic signs.

3. Colour Representation Comparison

The following analysis evaluates key color spaces evaluated during preprocessing design:

- **RGB:** Highly sensitive to illumination variations, rendering it less robust for real-world environmental shifts.
- **HSV:** Provides excellent color isolation by decoupling Hue from Saturation and Value, allowing robust detection under shadows.
- **YCrCb:** Separates luminance (Y) from chrominance components (Cr, Cb), making it ideal for luminance equalization routines.
- **LAB:** Offers broad perceptual uniformity, though computation is noticeably higher compared to HSV transformations.

4. Installation & Environment Setup

Follow these CLI commands to configure the development and deployment environment:

```shell
git clone https://github.com/autonav/tsr-preprocessing.git
cd tsr-preprocessing
python3 -m venv venv
source venv/bin/activate
pip install --upgrade pip
pip install -r requirements.txt
```

5. CLI Usage & Execution

Run the preprocessing pipeline on video feeds or image directories using the command-line interface:

```shell
python preprocess.py --input ./data/test_feed.mp4 --color-space HSV --clahe
--output ./processed/
```

6. Experimental Results & Performance Benchmarks

Key operational benchmarks across varied environmental conditions:

- **Daylight Clear Conditions:** Detection Accuracy of 98.2%, Processing Speed of 45 FPS, and Processing Latency of 22.2 ms.
- **Low Light / Shadow Conditions:** Detection Accuracy of 94.5%, Processing Speed of 42 FPS, and Processing Latency of 23.8 ms.
- **Heavy Rain / Adverse Weather:** Detection Accuracy of 91.8%, Processing Speed of 38 FPS, and Processing Latency of 26.3 ms.